SolarMax

Testing the Solar Panel with the AM33D Multimeter

Comparing panel output with and without tracking, using only the multimeter.

Contents

1. Overview

The panel can be characterized with the AM33D multimeter alone, using two readings on the bare panel:

- **Voc**: the open-circuit **voltage** (only the meter connected).
- **Isc**: the short-circuit **current** (meter placed across the panel).

Note

Isc and sunlight: a panel's short-circuit current (Isc) is proportional to the amount of sunlight reaching it, so it varies with how squarely the panel faces the sun. The open-circuit voltage (Voc) changes only slightly with orientation.

The panel is a Renogy 50 W (RNG-50D-SS). Typical full-sun values are **Voc $\approx$ 22 V** and **Isc up to $\approx$ 3 A**. Approximate power: $P \approx 0.74 \times V_{oc} \times I_{sc}$.

2. Safety and setup

- Take readings outside in steady, strong sun (clouds change the reading).
- The panel does not connect to the power supply; it generates its own power from light. Only the panel and the meter are involved.

Important

Shorting a solar panel is safe (unlike a battery): it can only supply its Isc, about 3 A. Two AM33D limits apply:

- For current, use the **10 A jack** with the **10 A dial position**. The `VΩmA` jack is fused at 250 mA, and 3 A would blow that fuse.
- The datasheet limits the 10 A range to **15 seconds at a time**. Take the Isc reading, then remove the probes.

3. AM33D quick settings

The black lead stays in **COM**. Two meter setups are used:

Reading	Dial position	Red lead jack	Typical display
Voc (voltage)	**200** in the DC-V block (V with =)	`VΩmA`	`21.8` (volts)
Isc (current)	**10A** in the DC-A block	`10A`	`2.85` (amps)

Note

Use the **DC** side (the "V" with a solid line above a dashed line, $V =$), not the AC "V∼". A leading **"−"** on the display means the leads are reversed; the number is still correct. `OL` indicates an over-range setting.

4. Measurement 1: Open-circuit voltage (Voc)

1. Red lead in the `VΩmA` jack, black lead in `COM`.
2. Turn the dial to **200** in the **DC voltage** block.
3. Touch the red probe to the panel's + wire and black to −.
4. Read the voltage. In full sun it is typically around **18 to 22 V**.

Tip

Press **HOLD** to freeze a reading for recording. Voc changes only slightly with orientation.

5. Measurement 2: Short-circuit current (Isc)

1. Move the red lead to the **10 A** jack (black stays in `COM`).
2. Turn the dial to **10A** in the **DC current** block.
3. Touch the probes across the panel's + and − wires. This briefly shorts the panel through the meter, which is acceptable for a PV panel.
4. Read the current within a few seconds, then lift the probes. In full sun with the panel aimed at the sun, values reach about **3 A**.

Important

Keep each Isc reading under 15 seconds. After measuring current, move the red lead back to the `VΩmA` jack, so a later probe touch does not short a circuit while the meter is in high-current mode.

6. Procedure: tracking vs. not tracking

Take both readings twice, back-to-back, in the same sunlight:

1. **Tracking:** panel aimed straight at the sun (tracker running and pointed at it, or aimed by hand so its face is square to the sun). Record Voc, then Isc.
2. **Not tracking:** panel left at a fixed, off-sun angle (for example flat/horizontal, or rotated about 45° away). Record Voc, then Isc.
3. Repeat the pair at several times of day (morning, noon, afternoon).

Time	**Tracking**		**Not tracking**		**Isc gain**
	Voc (V)	Isc (A)	Voc (V)	Isc (A)	(%)
Morning					
Noon					
Afternoon					

7. Interpreting the readings

- **Isc gain** $= \dfrac{I_{sc}(\text{tracking}) - I_{sc}(\text{not})}{I_{sc}(\text{not})} \times 100\%$. This expresses the difference in current between the two cases.
- **Approximate power** for any reading: $P \approx 0.74 \times V_{oc} \times I_{sc}$. Compute it for the tracking and not-tracking cases at each time.
- Voc typically changes little between the two cases, while Isc and power vary with orientation. Record and compare the values.

Note

For a consistent comparison, take each tracking / not-tracking pair quickly and in the same light. If a cloud passes during a pair, repeat that pair.

8. Troubleshooting

- **Current reads `0` or blows a fuse:** the `VΩmA` jack was used for current. Use the **10 A** jack and the **10A** dial position.
- **Voltage shows `OL`:** the range is too low. DC **200** is correct for about 22 V. Confirm the dial is on DC, not AC.
- **Leading "–" sign:** the leads are reversed. Swap them, or record the value as shown.

- **Readings vary:** passing clouds, or the panel warming or partially shaded. Wait for steady sun.
- **Meter turns off by itself:** auto power-off after 15 min. Turn it back on.

Meter details: `datasheets/ABMUAM33DRD-EN-v1.pdf` (AstroAI AM33D).